Calculus II Assignment 4

Part I. Regular Problems

1. Use the definition of a Taylor series to find the first four nonzero terms of the series for $f(x)$ centered at the given value of a.

   (a) $f(x) = xe^x, \quad a = 0$

   (b) $f(x) = \sqrt[3]{x}, \quad a = 8$

2. Obtain the Maclaurin series for the function

$$f(x) = \begin{cases} \frac{x - \sin x}{x^3} & \text{if } x \neq 0 \\ \frac{1}{6} & \text{if } x = 0 \end{cases}$$

   Hint:

$$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots, \quad \text{where } R = \infty$$

3. Consider

$$f(x) = \ln(1 + x).$$

   (a) Find the Maclaurin series for f using the definition of a Maclaurin series.

   (b) Find the associated radius of convergence and interval of convergence for the Maclaurin series.

   (c) Show that $f(x)$ is equal to the sum of the Maclaurin series for every x in the interval of convergence.

4. Use series obtained in Q3 to evaluate the limit:

$$\lim_{x \to 0} \frac{x - \ln(1 + x)}{x^2}$$

Part II. Group Problems

1. Find the Taylor polynomial of degree 3 with center 0 for f and estimate the remainder when $x \in [-1/2, 1/2]$:

   (a) $f(x) = 1 + x + x^2 + x^3 + x^4$.

   (b) $\cos(3x)$.

   (c) $\tan^{-1}(x)$.

2. Use a known Taylor series to calculate the value of the limit.

   (a) $\lim_{x\to 0} \frac{\sin x^2 - x^2}{x^6}$

   (b) $\lim_{x\to 0} \frac{e^{-2x} - 1}{x}$

   (c) $\lim_{x\to 0} \frac{\cos x^2 - 1}{x^4}$